DAA (LAB)

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EXPERIMENT :- **Huffman_coding(DA 1)**

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QUES:- SOLVE THIS QUESTION AND DRAW THE HUFFMAN TREE.

A	C	E	F	G	H	I	L	N	O	R	S	T	U	V	W	X	Y	DZ
3	3	26	5	3	8	13	2	16	9	6	27	22	2	5	8	4	5	3

<u>1. Given symbols and frequencies with codes:-</u>

Given symbol and frequencies :-

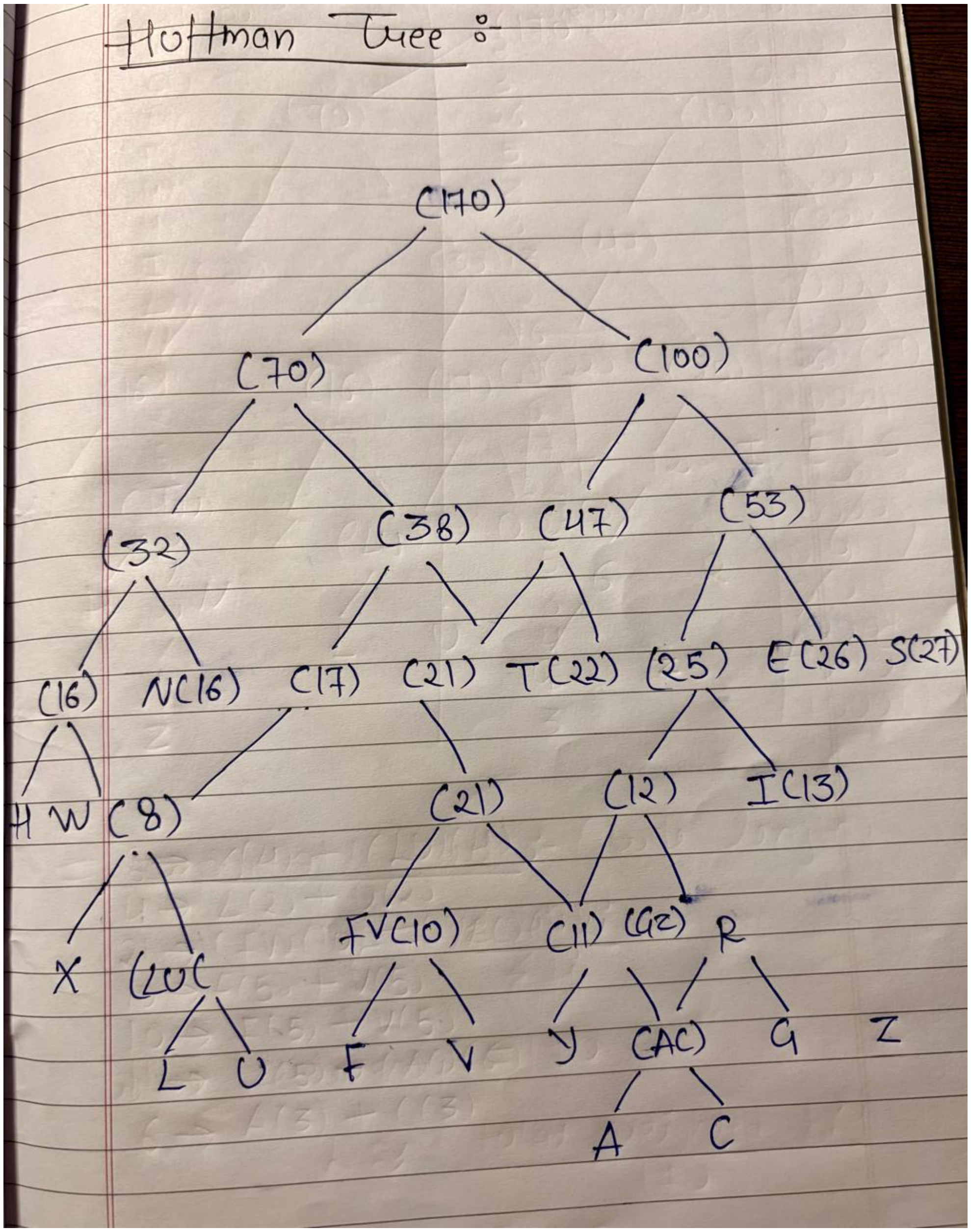
Symbol	frequencies	Code
A	3	011100
C	3	011101
E	26	1110
F	5	01010
G	3	100000
H	8	01000
I	13	101
L	2	00000
N	16	1100
O	9	001
R	6	10001
S	27	1111
T	22	1101
U	2	00001
V	5	01011
W	8	01011
X	4	0001
Y	5	0110
Z	3	100001

2. Sort symbols in ascending order of frequency:-

Sort Symbols in Ascending order of frequencies :-

L(2) , U(2)
A(3) , C(3) , G(3) , Z(3)
X(4)
F(5) , V(5) , Y(5)
R(6)
H(8) , W(8)
O(9)
I(13)
N(16)
T(22)
E(26)
S(27)

3.Huffman Tree (Optimal Merge Pattern):-



4.Verify total encoded message size:-

Important part :- (code length)

$$L = \frac{\sum f_i \cdot l_i}{\sum f_i}$$

Total frequency = 170

$$L \approx \frac{(\text{total weighted bits})}{170}$$

$$= \frac{720}{170} \approx 4.24 \text{ Bits/symbol}$$

5.explanation part:-

Explanation :-

final Merge = 170

⇒ Right Subtree :-

53 → E(26) + S(27)
47 → T(22) + 25
25 → 12 + I(13)
12 → (GZ)(6) + R(6)
6 → G(3) + Z(3)

2) Left Subtree :-

32 → N(16) + (HW)(16)
16 → H(8) + W(8)
38 → 17 + 21
17 → 8 + O(9)

$8 \rightarrow X(4) + (LU)(4)$

$4 \rightarrow L(2) + U(2)$

$21 \rightarrow (FV)(10) + (YAC)(11)$

$10 \rightarrow F(5) + V(5)$

$10 \rightarrow F(5) + V(5)$

$11 \rightarrow Y(5) + (AC)(6)$

$6 \rightarrow A(3) + C(3)$